Kartikeya Swarup Sharma

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LinkedIn / Portfolio / GitHub: https://github.com/kartikeyaswarup-m1

Experience

EY (AI & Data Intern) Gurugram, Haryana, India
Tech Consulting | CNS – AI & Data Sep 2025 – Dec 2025

- Built a **RAG-based** vendor security and compliance analysis system using **FastAPI**, PyMuPDF, Gemini embeddings, **Qdrant**, and Gemini LLMs to process and analyze multiple vendor documents.
- Implemented **semantic retrieval** and vendor-scoped vector search, retrieving relevant document evidence for each security control and preserving document type and clause-level metadata for traceable findings.
- Developed an **LLM-based compliance evaluation pipeline** that classified **22 security and compliance controls** as Covered, Partial, or Missing, with confidence scores and supporting evidence.
- Engineered a weighted risk-scoring system combining control classification, LLM confidence, and control weights to generate a **0–100 vendor risk score**, with findings and recommended remediation actions.
- Identified high-risk compliance gaps across multiple vendors through semantic document retrieval.
- **Developed 10 FastAPI endpoints** covering document upload, analysis, control/framework management, analysis history, and JSON/CSV report export, with **Dockerized backend** and Qdrant deployment.

Skills Summary

- **Programming Languages:** Python, Java, C, C++, JavaScript, SQL
- **AI/ML:** LLMs, RAG, Embeddings, Semantic Search, Vector Search, NLP, OpenCV
- **Frameworks:** FastAPI, Next.js, React.js, Tailwind CSS
- **Databases/Storage:** Firebase, Qdrant, FAISS, MySQL
- **Tools:** Docker, Git, GitHub, Linux, Postman
- **Certifications:** [NPTEL] Affective Computing , [SWAYAM] Introduction to Cyber Security , [NPTEL] Google Cloud Computing Foundations
- **Problem Solving:** Solved 150+ problems on LeetCode, Data Structures and Algorithms, Dynamic Programming,

Projects

RagTube — RAG-Based YouTube Q&A System

- Built a **full-stack Retrieval-Augmented Generation application** using **FastAPI, React, FAISS, Hugging Face embeddings, and Groq LLMs** for semantic question answering over YouTube video transcripts.
- Implemented an end-to-end **transcript ingestion and retrieval pipeline** with YouTube caption extraction, timestamp-aware chunking, embedding generation, FAISS indexing, and semantic top-k retrieval.
- Developed **streaming LLM responses over NDJSON**, grounding answers exclusively in retrieved transcript context and attaching timestamp references to relevant information.
- **Containerized** the application and made it **deployment-ready with Docker**, documented deployment to **Render + Cloudflare Pages**, and implemented provider configuration, CORS, health checks, and transcript fallbacks.

Face Recognition Attendance System

- Built a **real-time face recognition attendance system** using Python and OpenCV that detects faces from a live webcam feed and matches them against pre-generated facial encodings.
- Implemented **facial encoding generation and identity matching** using the face_recognition library, with optimized frame processing through image resizing and face-distance comparison.
- Integrated **Firebase Realtime Database and Cloud Storage** to retrieve student records, display profile information, and persist attendance counts and timestamps.
- Added **duplicate-attendance prevention using time-based validation**, updating a student's attendance only when the previous attendance was more than 30 seconds earlier.

Education

Bachelor of Technology in Computer Science & Engineering **(B.Tech. CSE)** CGPA: 7.17
Graphic Era Hill University, Dehradun, Uttarakhand, India Aug 2022 – Jul 2026